2.4 Decomposition in Noetherian and Distributive Lattices

An analogue of irreducible factorization in rings (see the Factorisation section) applies to Noetherian Lattices. Furthermore uniqueness holds when the lattice is distributive. For a canonical reference see [Bir40].

Definition 2.4.1 (Meet-Prime and Meet-Irreducible)
Let $(X, \leq)$ be a lattice and $x \in X$. Then we say that x is

- ***meet-irreducible*** *if $y \wedge z = x \implies y = x$ or $z = x$*
- ***meet-prime*** *if $y \wedge z \leq x \implies y \leq x$ or $z \leq x$*
- ***join-irreducible*** *if $y \vee z = x \implies y = x$ or $y = z$*
- ***join-prime*** *if $x \leq y \vee z \implies x \leq y$ or $x \leq z$*

The following result is proven in [Bir40] Ch. IX Lemma 4.1.

Proposition 2.4.2 (Prime = Irreducible)
Let $(X, \leq)$ be a lattice. In general meet-prime $\implies$ meet-irreducible and join-prime $\implies$ join-irreducible. If X is a distributive lattice, then the converse holds.

In the distributive case we denote by $\mathcal{M}(X)$ and $\mathcal{J}(X)$ the sub-poset of meet-prime and join-prime elements respectively.

Proof. The first statement is straightforward. Conversely suppose X is a distributive lattice and x is meet-irreducible. If $y \wedge z \leq x$ then $x = x \vee (y \wedge z) = (y \vee x) \wedge (z \vee x) \implies x = y \vee x$ or $x = z \vee x$, whence the result follows. ∎

Proposition 2.4.3
Let $(X, \leq)$ be a distributive lattice and Y a sub-lattice, then

$$\mathcal{M}(Y) = \mathcal{M}(X) \cap Y$$

$$\mathcal{J}(Y) = \mathcal{J}(X) \cap Y$$

In particular this holds when $Y = \{x\}^{\uparrow}, \{y\}^{\downarrow}, \{x\}^{\uparrow} \cap \{y\}^{\downarrow}$.

Proposition 2.4.4
Let $(X, \leq)$ be a distributive lattice. If it is chain-complete (resp. co-chain-complete) then so is $\mathcal{J}(X)$ (resp. $\mathcal{M}(X)$).

Every join-prime element is bounded above by a maximal join-prime element, and every meet-prime element is bounded below by a minimal meet-prime element

Proof. Let C be a chain of join-prime elements and $x := \bigvee C$. Suppose that $y \vee z \geq x$ then $y \vee z \geq w$ for all $w \in C$. Then $y \geq w$ or $z \geq w$ for all $w \in C$. Let $C_1 := \{w \in C \mid y \geq w\}$. If $C_1 = C$ then we are done as $x \leq y$. Otherwise suppose $w_0 \notin C_1$ then by prime-ness $z \geq w_0$. Clearly $w \leq w_0 \implies w \leq z$. Further $w \geq w_0 \implies w \not\leq y$ (as otherwise $w_0 \leq y$) whence $w \leq z$. Therefore $x \leq z$.

The last statement follows from Zorn's Lemma by considering the sub-lattices $\{x\}^{\uparrow}$ and $\{y\}^{\downarrow}$ which inherit the chain complete properties. ∎

Definition 2.4.5
Let $(X, \leq)$ be a lattice and $Y \subseteq X$ a finite subset. Then we say that Y is

- ***(meet-)irredundant*** *if no proper subset has the same meet*
- ***incomparable*** *(or an **antichain**) if no two elements are comparable*

Lemma 2.4.6 (incomparable $\iff$ irredundant)
Let $(X, \leq)$ be a lattice and $Y \subseteq X$ then Y meet-irredundant $\implies$ Y incomparable. Conversely if Y is a finite incomparable subset of meet-prime elements then Y is meet-irredundant.

Proof. The first part is straightforward, for if $y_1 \lneq y_2$ are elements of Y then $\bigwedge Y = \bigwedge Y \setminus \{y_2\}$.

Conversely suppose $Y' \subsetneq Y$ is such that $\bigwedge Y' = \bigwedge Y$. Choose $y_2 \in Y \setminus Y'$, then $\bigwedge Y' \leq y_2$, whence by definition of meet-prime (and induction) $y_1 \leq y_2$ for some $y_1 \in Y'$. ∎

The following is [Bir40] Chapter IX Theorem 9

Proposition 2.4.7 (Decomposition in Noetherian Lattice)
Let $(X, \leq)$ be a Noetherian distributive lattice. Then every element $x \in X$ has a unique decomposition

$$x = x_1 \wedge ... \wedge x_n$$

where x_i are meet-prime and meet-irredundant (equivalently incomparable). These are precisely the meet-primes minimal over x.

Dually, if $(X, \leq)$ is an Artinian distributive lattice, then every element $x \in X$ has a unique decomposition

$$x = x_1 \vee ... \vee x_n$$

where x_i are join-prime and join-irredundant (equivalently incomparable). These are precisely the join-primes maximal below x.

Proof. Let Y be the subset of elements which are not finite meets of meet-prime elements, and suppose it is non-empty. Then by (2.1.63) Y has a maximal element x_0. It cannot be meet-prime, and therefore not meet-irreducible (2.4.2), so there must exist elements $y_0, z_0 \in X$ such that $x_0 = y_0 \wedge z_0$ but $x_0 \lneq y_0$ and $x_0 \lneq z_0$. By maximality y_0, z_0 are finite meets of prime elements, and therefore so is x_0 a contradiction.

Therefore we have a decomposition into distinct primes

$$x = x_1 \wedge ... \wedge x_n$$

Consider the family of subsets of $\{x_1, ..., x_n\}$ which have the same meet. Then there exists a minimal subset which by definition is meet-irredundant and by (2.4.6) incomparable. Suppose there is another such decomposition $x = x'_1 \wedge ... \wedge x'_m$. Then for every $i = 1...n$ we have $x'_{\sigma(i)} \leq x_i$ and for every $j = 1...m$ we have $x_{\tau(j)} \leq x'_j$ whence $x_{\tau(\sigma(i))} \leq x'_{\sigma(i)} \leq x_i$. As the decomposition is incomparable we have $\tau(\sigma(i)) = i$ and $x'_{\sigma(i)} = x_i$. Therefore σ is injective and $n \leq m$. By symmetry $m \leq n$ and σ is a bijection. In otherwords the decomposition is unique.

Note $x \leq z$ and z meet-prime implies $x_j \leq z$ for some j. Therefore if z is a minimal prime then $x_j = z$ and all the meet-primes minimal over x must occur in the decomposition. Similarly if $z \leq x_i$ then by incomparability $x_j = z = x_i$. Therefore each x_i is also minimal. ∎